Owen's Math Study Guide

Name: _____ Date: _____

Simple Solutions Standards-Based Mathematics · Lesson #27 (book page 54) · Grade 7

How to use this packet

1. Read the strategy box for a skill. Try its example on scrap paper before reading the answer.
2. Go through the worked solutions for Lesson #27. Cover the steps and see if you can do each one yourself first.
3. Do the extra practice pages (A–E). Show your work: the steps count as much as the answer.
4. Check with the answer key at the back. For anything you missed, re-read that strategy box and try one more.
5. Bonus: open Owen_Lesson27_Practice.html on a computer or tablet for a quiz that talks and checks your answers.

What's in Lesson #27

1. Integers & negative numbers	#1, #3	2. Surface area of a rectangular prism ★	#2
3. Combining like terms	#4	4. Fractions: adding & tenths ★	#5, #7
5. Proportional relationships on a graph	#6	6. Percent of a number ★	#8
7. Writing expressions ★	#9	8. Unit rate	#10
9. Dividing decimals	#11	10. Inequalities ★	#12

★ = focus skill (marked on the page in pencil). These get extra practice.

Focus problems

Pencil marks on the page show a checkmark at #2 and circles at #8, #9, and #12 (maybe a mark at #7 too). This packet gives extra practice on:

• Surface area of a box (finding a missing side) • Percent of a number • Tenths
• Writing an expression from words • Writing and solving inequalities

Two problems we couldn't fully see

#3: the multiple-choice answers are cut off in the photo. We teach the skill (5 + (−2) = 5 − 2 = 3), so you can pick the matching choice in your book.

#6: the graph isn't in the photo. We teach the rule (a straight line through (0, 0)) and give you our own graphs to practice on.

Strategies by Skill

Each box: how to do it, an example, and something to remember.

1. Integers & negative numbers

Lesson #1, #3

- Money you have is positive. Money you owe is negative. A balance below $0 means you owe.
- Adding a negative number means moving LEFT on the number line. So $a + (-b)$ is the same as $a - b$.
- Subtracting from 0: $0 - 2.50 = -2.50$.

Example: Your account is at $0 and you're charged $2 a day for 3 days: $3 \times 2 = 6$, and $0 - 6 = -6$, so you owe $6. $5 + (-2)$: start at 5 and move 2 left to land on 3. That's the same as $5 - 2$.

Remember: Plus a negative is the same as minus.

2. Surface area of a rectangular prism ★ FOCUS

Lesson #2

- A box has 6 faces in 3 matching pairs: top & bottom (l × w), front & back (l × h), left & right (w × h).
- Surface area: SA = 2(lw + lh + wh). The answer is in square units (m^2, cm^2, in^2).
- Missing side? Call it x and fill in the formula. Then undo it step by step: divide by 2, subtract the face you know, and divide by what's left (the two known sides added together).

Example: Box 5 × 4 × 3: SA = 2(5·4 + 5·3 + 4·3) = 2(20 + 15 + 12) = 2(47) = 94 square units.

Remember: Check a missing-side answer by plugging it back into the formula.

3. Combining like terms

Lesson #4

- Like terms have the same variable part: 4n and −3n are like terms, and so are the plain numbers 9 and −6.
- Each term keeps the sign in front of it. Combine variables with variables and numbers with numbers.
- Write the variable term first. 1n is written as just n.

Example: $7x + 4 - 2x - 1 \rightarrow (7x - 2x) + (4 - 1) = 5x + 3$.

Remember: Circle the x terms, box the plain numbers, then combine each group.

4. Adding fractions; fractions as tenths

Lesson #5, #7

- To add fractions, get a common denominator. Change mixed numbers to improper fractions (1¾ = 7/4), or add the whole numbers and the fractions separately.
- Add the numerators and keep the denominator. Simplify, then change back to a mixed number.
- "How many tenths?" Make the denominator 10: multiply the top and bottom by the same number.

Example: ½ + 1¼ = 2/4 + 5/4 = 7/4 = 1¾. 2/5 = 4/10, so 2/5 is 4 tenths (0.4).

Remember: A whole number is a fraction too: 2 = 8/4.

5. Proportional relationships on a graph

Lesson #6

- Proportional means $y = k \cdot x$. Every y is the same number (k) times its x.
- On a graph it must be BOTH a straight line AND pass through the origin (0, 0).
- NOT proportional: a line that starts above or below (0, 0), or a graph that curves.

Example: The points (1, 3), (2, 6), (3, 9) make a straight line through (0, 0), so they're proportional (k = 3). A taxi that costs $2 just to get in, plus $1 per mile, starts at (0, 2). Not proportional!

Remember: Straight + through zero = proportional.

Strategies by Skill (continued)

6. Percent of a number ★ FOCUS

Lesson #8

- Percent means "out of 100": 35% = 35/100 = 0.35.
- "Of" means multiply: 35% of 300 = 0.35 × 300.
- Mental math: 10% means moving the decimal 1 place left. 5% is half of 10%, and 1% means moving it 2 places left. Build any percent from those.

Example: 15% of 80: 10% is 8 and 5% is 4, so 8 + 4 = 12. Or 0.15 × 80 = 12.

Remember: Check if it makes sense: 50% is half, 25% is a quarter.

7. Writing expressions ★ FOCUS

Lesson #9

- Pick a letter (a variable) for the unknown amount and say what it means: "Let ℓ = the length."
- Add (+): more than, increased by, plus, total. Subtract (−): less than, decreased by, fewer, minus.
- Multiply (×): times, twice, triple. Divide (÷): half of, split equally.
- Careful: "5 less than p" is p − 5, not 5 − p. You start with p and take away 5.

Example: "A ladder is 4 feet longer than a table (t)" → t + 4. "Twice Ben's age (b)" → 2b.

Remember: If the problem doesn't give a letter, pick one and say what it stands for.

8. Unit rate

Lesson #10

- A unit rate is the amount for ONE: cost per bag, miles per hour, and so on.
- Divide the total by the number of units: total cost ÷ number of items.
- Dividing by 10 moves the decimal point 1 place to the left.

Example: $14.40 for 6 notebooks: 14.40 ÷ 6 = $2.40 per notebook.

Remember: Money answers get 2 decimal places: $3.3 → $3.30.

9. Dividing decimals

Lesson #11

- Make the divisor (the number you divide BY) a whole number by moving its decimal point to the right.
- Move the decimal in the dividend (the first number) the SAME number of places. Add zeros if you need to.
- Then divide like normal. Check by multiplying.

Example: 4.8 ÷ 0.06: move both 2 places, so it becomes 480 ÷ 6 = 80. Check: 80 × 0.06 = 4.8 ✓

Remember: Dividing by a number less than 1 makes the answer bigger.

10. Inequalities ★ FOCUS

Lesson #12

- Words to symbols: "is greater than" >, "is less than" <, "at least" ≥, "at most / no more than" ≤.
- Solve it like an equation: do the same thing to both sides to get the variable by itself.
- The answer is a whole range of numbers. w > 4 means any number bigger than 4 (4 itself doesn't count).
- Graph: open circle for > or <, closed (filled) circle for ≥ or ≤. Shade toward the numbers that work.

Example: "A number (k) minus 3 is less than 10" → k − 3 < 10 → add 3 to both sides → k < 13.

Remember: Test your answer: pick a number in your range and plug it in.

Lesson #27: Worked Solutions

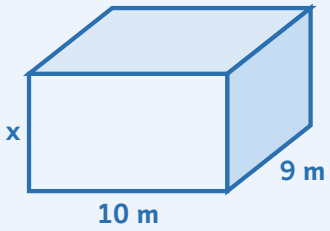
Try each problem first, then check the steps. A gold star means it's a focus problem.

1 **Mario started with a zero balance on his library account. A book he borrowed was 7 days overdue, and he was charged $0.25 per day. Explain the current status of Mario's account.**

Total fine: 7 days × $0.25 = $1.75.
Start at $0 and take away $1.75: 0 − 1.75 = −1.75.
A negative balance means money is owed.

Answer: His balance is −$1.75. Mario owes the library $1.75.

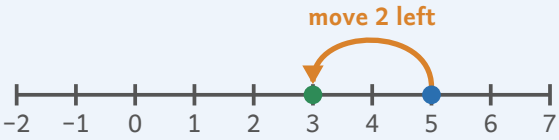
2 ★ **The surface area of a rectangular prism is 408 m². Two of the dimensions are 9 m and 10 m. Find the measurement of the other dimension.**



SA = 2(lw + lh + wh). Call the missing side x.
2(9·10 + 9x + 10x) = 408
2(90 + 19x) = 408 → divide by 2
90 + 19x = 204 → subtract 90
19x = 114 → divide by 19
x = 6
Check: 2(90 + 54 + 60) = 2(204) = 408 ✓

Answer: The other dimension is 6 m.

3 **Use the number line to solve 5 + (−2). Give the sum. Which of these has the same value?**



Start at 5. Adding a negative means moving LEFT.
Move 2 to the left and you land on 3.
Adding −2 is the same as subtracting 2: 5 + (−2) = 5 − 2.

Answer: The sum is 3. It has the same value as 5 − 2.

Note: the answer choices are cut off in the photo. Pick the choice in your book that equals 3 (like 5 − 2).

4 **Simplify this expression. 4n + 9 − 6 − 3n**

Group the n terms: 4n − 3n = 1n = n.
Group the numbers: 9 − 6 = 3.
Write the variable term first.

Answer: n + 3

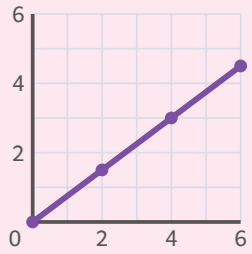
Lesson #27: Worked Solutions (continued)

5 **Charlotte rode ¼ mile to her grandmother's house, 1¾ mile to the library, 2 miles to the park, and ½ mile to her home. How far did Charlotte ride?**

Change everything to fourths: ¼ = 1/4, 1¾ = 7/4, 2 = 8/4, ½ = 2/4.
Add: 1/4 + 7/4 + 8/4 + 2/4 = 18/4.
Simplify: 18/4 = 9/2 = 4½.
(Shortcut: ¼ + ¾ = 1 whole, so 1 + 1 + 2 + ½ = 4½.)

Answer: Charlotte rode 4½ miles (that's 9/2, or 4.5 miles).

6 **Study the graph. Does it represent a proportional relationship? How do you know?**



Check two things:
1) Is it a straight line?
2) Does it go through the origin (0, 0)?
If BOTH are yes, it's proportional. If either is no, it's not.
Example at right: a straight line through (0, 0), so it's proportional.

Answer: It's proportional only if the graph is a straight line through (0, 0). Say which part is true or false for your graph.

Note: the graph for #6 isn't in the photo, so we can't answer Owen's exact graph. Use the two-question test on the graph in your book. There's more practice in Practice E.

7 **How many tenths of a kilometer are in ⅗ of a kilometer?**

Tenths means the denominator should be 10.
★ 5 × 2 = 10, so multiply the top and bottom by 2: 3/5 = 6/10.

Answer: 6 tenths (0.6 km).

8 **If 35% of 300 children had zero cavities, how many children was that?**

35% = 35/100 = 0.35. "Of" means multiply.
★ 0.35 × 300 = 105.
Mental math: 10% of 300 = 30, so 30% = 90. 5% = 15. 90 + 15 = 105.

Answer: 105 children.

Lesson #27: Worked Solutions (continued)

9 **The height of the book shelf is 3 feet more than its length. Write an expression to represent the height of the book shelf.**

★ The problem doesn't give a letter, so pick one. Let ℓ = the length of the shelf (in feet).
"3 feet more than the length" means add 3 to the length.

Answer: Height = ℓ + 3 (L + 3, x + 3, or length + 3 also work)

Don't write 3ℓ. That means 3 TIMES the length. "More than" means add.

10 **Cal paid $33.00 for 10 bags of cotton candy. What was the unit rate in cost per bag?**

Unit rate = total ÷ number of bags.
$33.00 ÷ 10 = $3.30 (dividing by 10 moves the decimal 1 place left).

Answer: $3.30 per bag.

11 **3.6 ÷ 0.09 = ?**

Make the divisor a whole number: 0.09 → 9 (move the decimal 2 places right).
Move the decimal in 3.6 two places too: 3.6 → 360.
360 ÷ 9 = 40. Check: 40 × 0.09 = 3.6 ✓

Answer: 40

12 **Write and solve an inequality that means "4 plus a number (w) is greater than 8."**

★ "4 plus a number w" → 4 + w. "is greater than 8" → > 8.
Inequality: 4 + w > 8
Subtract 4 from both sides: w > 4.
Test: w = 5 → 4 + 5 = 9 > 8 ✓. w = 4 → 8 > 8? No, so 4 isn't included (open circle).



Answer: 4 + w > 8, so w > 4

Quick self-check: can you explain it out loud?

• Why does a negative balance mean Mario owes money? • Why do we multiply by 2 in the surface-area formula?
• Why is 4n − 3n just n? • What two things make a graph proportional? • What does "of" mean in "35% of 300"?
• Why is "3 more than ℓ" written ℓ + 3 and not 3ℓ? • Why doesn't w = 4 work in w > 4?

Practice A1: Surface Area of a Box

Name: _____ Date: _____

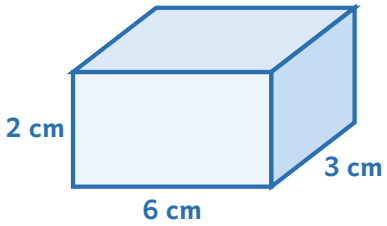
SA = 2(lw + lh + wh). Find the area of each pair of faces, add them, then double. Units are squared.

Face helper (use it for #1):

top & bottom: l × w = _____ × _____ = _____ front & back: l × h = _____ × _____ = _____

left & right: w × h = _____ × _____ = _____ SA = 2(_____ + _____ + _____) = 2(_____) = _____

1. Find the surface area of a box that is 6 cm long, 3 cm wide, and 2 cm tall.



Answer: _____

2. A cube has edges that are 4 inches long. What is its surface area?

Answer: _____

3. Find the surface area of a box that is 8 ft by 5 ft by 2 ft.

Answer: _____

4. A cereal box is 7 in long, 2 in wide, and 10 in tall. How much cardboard covers the outside?

Answer: _____

Practice A2: Find the Missing Dimension

Name: _____ Date: _____

Call the missing side x. Fill in SA = 2(lw + lh + wh), divide both sides by 2, subtract the face you know, then divide.

Fill in the steps: SA = 164 cm², two sides are 10 cm and 4 cm.

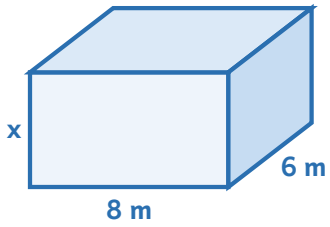
2(_____ · _____ + _____ x + _____ x) = _____

2(_____ + _____ x) = _____ → divide both sides by 2

_____ + _____ x = _____ → subtract

_____ x = _____ → x = _____ cm Check: 2(_____ + _____ + _____) = 164?

5. The surface area of a rectangular prism is 236 m². Two of the dimensions are 8 m and 6 m. Find the other dimension.



Answer: _____

6. The surface area of a rectangular prism is 174 cm². Two of the dimensions are 9 cm and 3 cm. Find the other dimension.

Answer: _____

7. The surface area of a rectangular prism is 320 in². Two of the dimensions are 12 in and 4 in. Find the other dimension.

Answer: _____

Practice B: Percent of a Number & Tenths

Name: _____ Date: _____

Percent: change the % to a decimal and multiply ("of" means ×), or build it from 10% and 5%. Tenths: make the denominator 10.

Percent of a number

1. 10% of 70 = ?

Answer: _____

2. 25% of 120 = ?

Answer: _____

3. 35% of 200 = ?

Answer: _____

4. 45% of 300 = ?

Answer: _____

5. At a school, 60% of the 400 students ride the bus. How many students ride the bus?

Answer: _____

6. A family leaves a 15% tip on a $40 restaurant bill. How much is the tip?

Answer: _____

7. Of the 250 kids at a camp, 12% are left-handed. How many kids is that?

Answer: _____

Fractions as tenths

8. How many tenths of a meter are in 2/5 of a meter?

Answer: _____

9. How many tenths of a kilometer are in 4/5 of a kilometer?

Answer: _____

10. How many tenths of a mile are in 1½ miles?

Answer: _____

11. Write ⅗ as a decimal.

Answer: _____

Practice C: Writing Expressions

Name: _____ Date: _____

Pick a letter for the unknown (if none is given). more than → +, less than → − ("4 less than ℓ" is $\ell - 4$, not $4 - \ell$), times → ×, half → ÷.

1. 5 more than a number n

Answer: _____

2. a number t decreased by 8

Answer: _____

3. 8 times a number k

Answer: _____

4. half of a number h

Answer: _____

5. The width of a poster is 4 inches less than its length (ℓ). Write an expression for the width.

Expression: _____

6. Maya has 3 times as many stickers as Leo. Leo has s stickers. Write an expression for Maya's stickers.

Expression: _____

7. A garden's length is 6 meters more than its width (w). Write an expression for the length.

Expression: _____

8. Sam is 2 years younger than his sister. Choose a variable for his sister's age, then write an expression for Sam's age.

Let ____ = _____

Expression: _____

9. A desk's height is 2 feet more than its width. Choose a variable for the width, then write an expression for the height.

Let ____ = _____

Expression: _____

Practice D: Inequalities

Name: _____ Date: _____

greater than >, less than <, at least ≥, at most ≤. Solve like an equation: do the same thing to both sides.

Write the inequality, then solve it

1. 7 plus a number (w) is greater than 12.

Inequality: _____

Solve: _____

2. A number (k) minus 4 is less than 9.

Inequality: _____

Solve: _____

3. A number (y) plus 6 is at least 10.

Inequality: _____

Solve: _____

4. 3 plus a number (p) is at most 10.

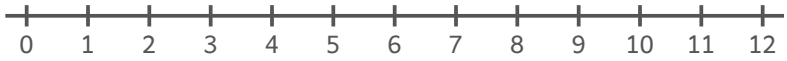
Inequality: _____

Solve: _____

Solve, then graph the answer on the number line

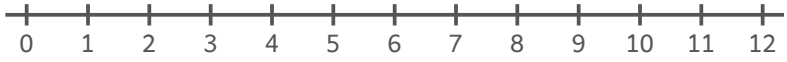
5. $x + 6 > 10$

Answer: _____



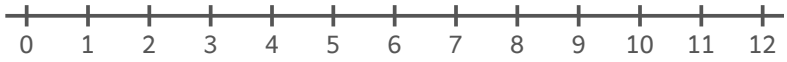
6. $m - 2 < 5$

Answer: _____



7. $n + 9 \geq 15$

Answer: _____



True or false? Test the number in the inequality.

8. Is w = 4 a solution of w > 4?

Yes / No: _____

9. Is w = 4.5 a solution of w > 4?

Yes / No: _____

10. Is y = 4 a solution of y ≥ 4?

Yes / No: _____

11. Is k = 13 a solution of k < 13?

Yes / No: _____

Practice E: Mixed Review

Name: _____ Date: _____

Integers, like terms, fractions, unit rate, dividing decimals, and proportional graphs.

1. Jada's lunch account starts at $0. She is charged $0.50 a day for 6 days. What is her balance? What does it mean?

Answer: _____

2. −4 + 7 = ?

Answer: _____

3. 6 + (−9) = ? Is it the same as 6 − 9?

Answer: _____

4. Simplify: 5x + 8 − 3 − 2x

Answer: _____

5. Simplify: 9a − 4 + 2a + 10

Answer: _____

6. Liam walked ½ mile, then 2¼ miles, then ¾ mile. How far did he walk in all?

Answer: _____

7. A pack of 5 juice boxes costs $18.75. What is the cost per juice box?

Answer: _____

8. 2.4 ÷ 0.08 = ?

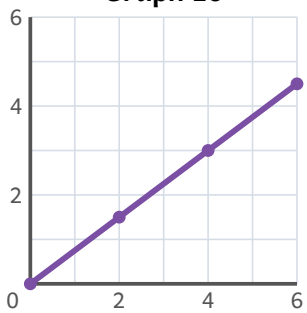
Answer: _____

9. 7.2 ÷ 0.9 = ?

Answer: _____

Proportional or not? (Is it a straight line through (0, 0)?)

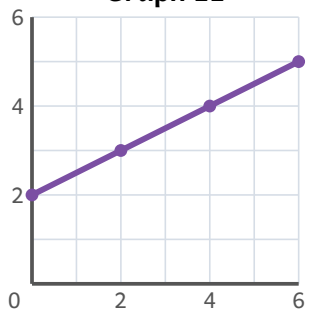
Graph 10



Proportional? Yes / No

Why? _____

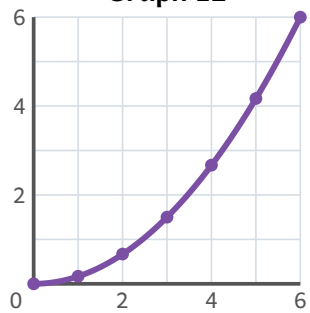
Graph 11



Proportional? Yes / No

Why? _____

Graph 12



Proportional? Yes / No

Why? _____

Answer Key

For checking after you finish. Lesson #27 answers first, then Practice A–E.

Lesson #27

#1 −$1.75: Mario owes the library $1.75 (7 × 0.25 = 1.75)
#2 6 m (2(90 + 19x) = 408 → 90 + 19x = 204 → 19x = 114 → x = 6)
#3 Sum = 3. Same value as 5 − 2. (The choices are cut off in the photo.)
#4 n + 3
#5 4½ miles (1/4 + 7/4 + 8/4 + 2/4 = 18/4 = 9/2)
#6 Proportional only if it's a straight line AND goes through (0, 0). (The graph isn't in the photo.)
#7 6 tenths (3/5 = 6/10)
#8 105 children (0.35 × 300)
#9 ℓ + 3 (any letter for the length, + 3)
#10 $3.30 per bag (33 ÷ 10)
#11 40 (360 ÷ 9)
#12 4 + w > 8 → w > 4

Practice A: Surface area

1. 72 cm^2 (2(6·3 + 6·2 + 3·2) = 2(18 + 12 + 6) = 2(36) = 72)
2. 96 in^2 (6 faces × (4 × 4) = 6 × 16 = 96)
3. 132 ft^2 (2(40 + 16 + 10) = 2(66) = 132)
4. 208 in^2 (2(14 + 70 + 20) = 2(104) = 208)
5. 5 m (2(48 + 8x + 6x) = 236 → 48 + 14x = 118 → 14x = 70 → x = 5)
6. 5 cm (2(27 + 9x + 3x) = 174 → 27 + 12x = 87 → 12x = 60 → x = 5)
7. 7 in (2(48 + 12x + 4x) = 320 → 48 + 16x = 160 → 16x = 112 → x = 7)
box Fill-in steps: 3 cm (2(10·4 + 10x + 4x) = 164 → 40 + 14x = 82 → 14x = 42 → x = 3)

Practice B: Percent & tenths

1. 7 (move the decimal 1 place left)
2. 30 (25% = ¼, and 120 ÷ 4 = 30)
3. 70 (0.35 × 200 = 70 (30% = 60, 5% = 10))
4. 135 (0.45 × 300 = 135 (40% = 120, 5% = 15))
5. 240 students (0.60 × 400 = 240)
6. $6.00 (10% = $4, 5% = $2, so $4 + $2 = $6)
7. 30 kids (0.12 × 250 = 30 (10% = 25, 2% = 5))
8. 4 tenths (2/5 = 4/10)
9. 8 tenths (4/5 = 8/10)
10. 15 tenths (1½ = 3/2 = 15/10)
11. 0.6 (3/5 = 6/10 = 0.6)

Practice C: Expressions

1. n + 5
2. t − 8
3. 8k

Answer Key (continued)

4. h ÷ 2 (or h/2)
5. ℓ − 4
6. 3s
7. w + 6
8. s − 2 (any letter − 2)
9. w + 2 (any letter + 2)

Practice D: Inequalities

1. 7 + w > 12 → w > 5
2. k − 4 < 9 → k < 13
3. y + 6 ≥ 10 → y ≥ 4
4. 3 + p ≤ 10 → p ≤ 7
5. x > 4 (open circle at 4, shade right)
6. m < 7 (open circle at 7, shade left)
7. n ≥ 6 (closed circle at 6, shade right)
8. No (4 > 4 is false)
9. Yes
10. Yes (≥ includes 4)
11. No

Practice E: Mixed review

1. −$3.00 (she owes $3.00) (6 × 0.50 = 3.00; 0 − 3.00 = −3.00)
2. 3 (start at −4, move 7 right)
3. −3; yes (adding −9 = subtracting 9)
4. 3x + 5 (5x − 2x = 3x; 8 − 3 = 5)
5. 11a + 6 (9a + 2a = 11a; −4 + 10 = 6)
6. 3½ miles (2/4 + 9/4 + 3/4 = 14/4 = 3½)
7. $3.75 (18.75 ÷ 5 = 3.75)
8. 30 (240 ÷ 8 = 30)
9. 8 (72 ÷ 9 = 8)

G10 Yes: a straight line through (0, 0)

G11 No: it's straight, but it starts at (0, 2), not (0, 0)

G12 No: it curves (not a straight line), even though it starts at (0, 0)

For grown-ups

• Focus skills (pencil marks on the page): surface area (#2), tenths (#7), percent (#8), expressions (#9), inequalities (#12).
• #3's choices and #6's graph weren't in the photo. Ask Owen to show you those two in his book and use the rules on the strategy pages.
• The interactive page (Owen_Lesson27_Practice.html) works offline, reads hints aloud, and lets him redo missed problems.